

School of Computer and Systems Sciences
Jawaharlal Nehru University
Mid Semester Examination, (Winter Semester 2025)

Database Management Systems (CS-407)

Duration: 90 Minutes

Max. Marks 25

Q1 A company is organised into departments. Each department has Employees working in it. Each department has a name and a number associated with it. Each employee has an employee number, name, date of birth, gender, date of joining, designation and basic pay. Each department is managed by a manager. There are also supervisors in each department who supervises a set of employees. Each department controls a number of projects. Each project has a code and a name. One project is controlled by one department only. An employee can work on any number of projects on a day. The date an employee worked. In time and out time has to be kept track. The company also keeps track of dependents of each employee. The attributes of dependents include name, date of birth, gender and relationship with employee.

- (a) Construct and ER diagram for the above scenario. (5)
- (b) Map the ER diagram into relations. (5)

OR

Q1 (a) Explain various components of DBMS? (4)

(b) What is Normalization ? Explain various normal forms with suitable examples in DBMS. (6)

Q2 Consider the following relational database schema: (6)

License(License_id, Name, Address, Phone, DOB)

Registration (License_id, Car_id, Registered) // value of Registered is : yes/ no

Car(Car_id, Model, Year, Color)

Write down the relational algebra expressions for the following queries:

- a) Find the name and address of the license holders who owns a white car.
- b) Find the name and address of car holders who have cars that are yet to be registered.
- c) Find the name of the car holders and model of the cars that they hold.
- d) Find the license_id and phone number of car holders who holds a car launched in 2021.

Q3 Let $R(A, B, C, D, E)$ be a relation and a given set of FDs $F = \{A \rightarrow D, BC \rightarrow AD, C \rightarrow B, E \rightarrow A, E \rightarrow D\}$ Find the minimal cover of F . (3)

Q4 Given a set of functional dependency $F = \{A \rightarrow B, B \rightarrow C, C \rightarrow D\}$, determine closure F^+ using Armstrong's axioms. (3)

Q5 Given a relation $R(A, B, C, D, E)$ and a set of functional dependency $F = \{A \rightarrow B, A \rightarrow C, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$, determine FD $BD \rightarrow CD$ is logically implied by F or not. Also determine the super keys of R that are not candidate keys. (3)

**School of Computer and Systems Sciences
Jawaharlal Nehru University
New Delhi-110067
Database Management Systems (CS-407)
End Semester Exam, Winter Semester – 2025**

Duration: 3 Hours

Max. Marks: 50

All questions are compulsory.

Q1 (a) Consider the relation $R = \{A, B, C, D, E, F, G, H\}$ and a set of FDs $S = \{CH \rightarrow G, A \rightarrow BC, B \rightarrow CFH, E \rightarrow A, F \rightarrow EG\}$. Determine the decomposition of R that satisfy 3NF.

(b) What is join dependency? Explain the consequences in the database with join dependency and how it can be addressed? (5x2=10)

Q2 Consider the following relations:

Salesman (sid, s_name, city, commission)

Customer (cid, cust_name, city, grade, sid)

Order (order_no, purchase_amount, order_date, cid, sid)

Write SQL expressions for the following queries:

- Find customers who are served by a salesperson and the salesperson earns commission between 10% and 15%.
- Display orders which exceed average value of orders placed on 31-March-2025.
- Count the number of customers with grades more than the average grades in New Delhi.
- Find the name and id of salesmen who have more than one customer.
- Find the salesman who deal with exactly one customer.

(2x5=10)

Q3 Consider the following schedules: S1 and S2:-

S1: R1(X) R3(Y) W1(X) W2(X) W2(Y) W1(Z) R3(Z)

S2: R1(X) R2(Y) W1(X) W3(X) W3(Y) W1(Z) R2(Z)

Which of the above given schedules are NOT allowed by Time Stamp Ordering Protocol? (Assume timestamps, $T_1 = 5, T_2 = 10, T_3 = 15$) (10)

Q4 Consider the following schedule-

S: R2(X)W3(X)W1(X)W2(Y)R2(Z)R4(X)R4(Y)

- Determine whether the schedule is conflict serializable or not?
- Determine whether the schedule is valid under 2PL or not? (5x2=10)

Q5 (a) Explain graph-based protocol for concurrency control and how they are different from 2PL?

- What is deadlock? Differentiate *wait-die* and *wound-wait* protocols for deadlock prevention. (5x2=10)